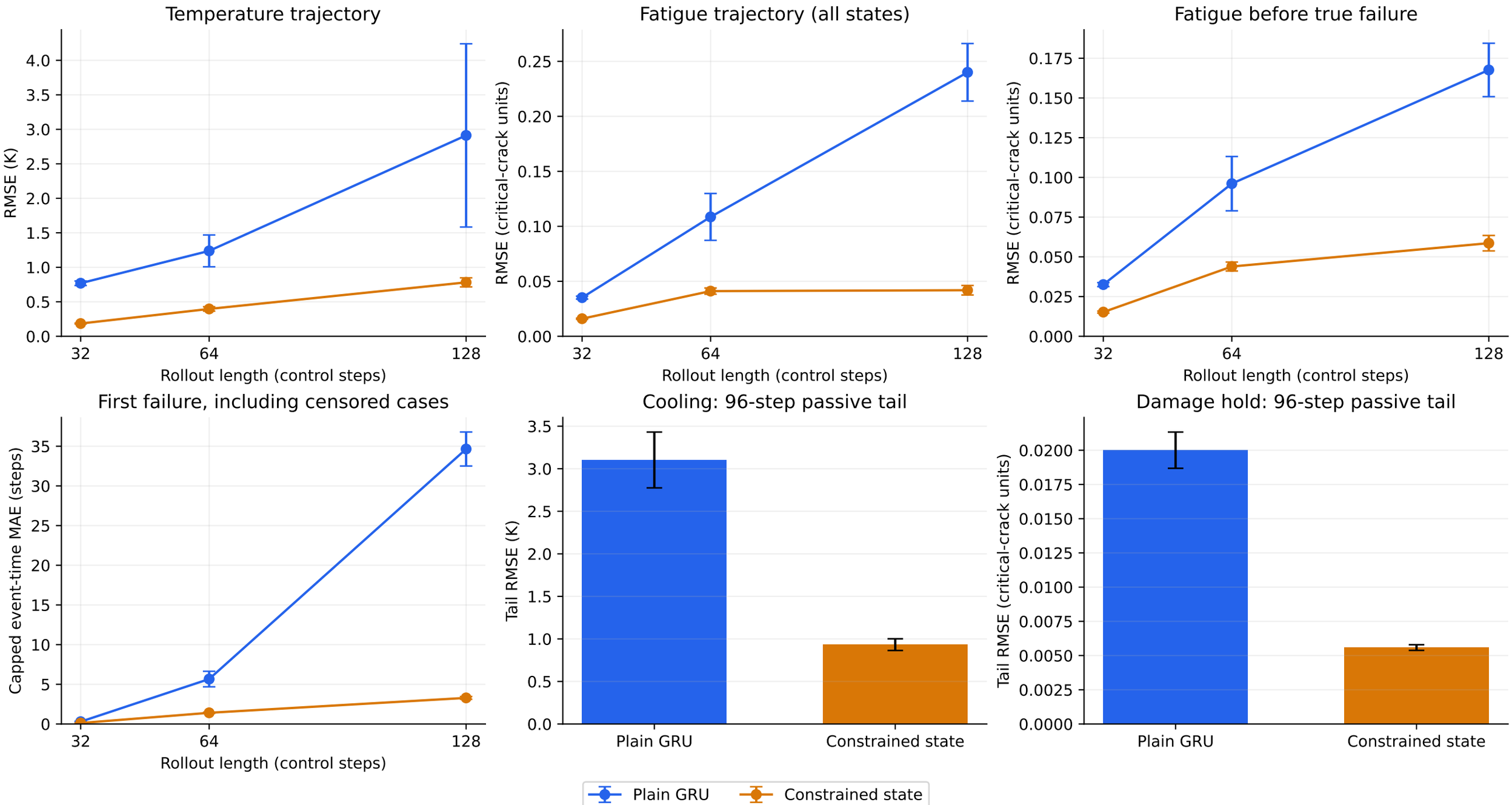
Local synthetic history experiment — 3 initialization seeds
Error bars: sample SD; training history length: 32 steps
Temperature trajectory
RMSE (K)
Rollout length (control steps)
Fatigue trajectory (all states)
RMSE (critical-crack units)
Rollout length (control steps)
Fatigue before true failure
RMSE (critical-crack units)
Rollout length (control steps)
First failure, including censored cases
Capped event-time MAE (steps)
Rollout length (control steps)
Cooling: 96-step passive tail
Tail RMSE (K)
Plain GRU
Constrained state
Damage hold: 96-step passive tail
Tail RMSE (critical-crack units)
Plain GRU
Constrained state
Plain GRU
Constrained state